

ANAMIKA RAJENDRAN

Las Cruces, New Mexico, USA

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EDUCATION

PhD in Computer Science

January 2025 - Expected December 2029

New Mexico State University, Las Cruces, New Mexico, USA

Advisor: Dr. Shiva Darian Co-Advisor: Dr. Bill Hamilton

CGPA: 4.00/4.00

Master of Science in Computer Science

January 2023 - December 2024

New Mexico State University, Las Cruces, New Mexico, USA

Advisor: Dr. Joshua Reynolds

CGPA: 3.90/4.00

Bachelor of Technology in Computer Science Engineering

August 2015 - January 2019

Kerala Technological University, Trivandrum, Kerala, India

CGPA: 8.90/10.00

RESEARCH EXPERIENCE

Data Advocacy Research (DAR) Lab, New Mexico State University

January 2025 - Present

Graduate Research Assistant, PhD

Las Cruces, New Mexico, USA

- *Collaborators:* Dr. Shiva Darian, Jonathan Leuenberger, Jessica Needle (University of Texas at Austin), Augusto Penzo Jara, Mason Eiland
- Built a local **WhisperX** transcription and diarization pipeline (fully on-device to preserve participant confidentiality) and integrated it with a topic-modeling tool and graphical interface to support large-scale qualitative analysis
- Conducted qualitative coding of 32 migrant interviews, contributing to the **CHI 2026** paper “Transient Non-Use: How People in Migration Experience Digital Disconnection”
- Investigating surveillance, privacy, and trust/verification practices among people on the move and analyzing the EOIR (Executive Office for Immigration Review) court dataset as part of ongoing PhD research

Wayllace’s AI for Social Impact (WASI) Lab, New Mexico State University

January 2025 - Present

Collaborating Graduate Research Assistant, PhD

Las Cruces, New Mexico, USA

- *Collaborators:* Dr. Christabel Wayllace, Manuel Mares Solano
- Designing a single-pointing on-screen **Augmentative and Alternative Communication (AAC)** keyboard that integrates a Large Language Model to help non-verbal users with severe motor impairments construct words and sentences
- Ran a year-long study with a long-time joystick-based AAC user showing approximately 60% fewer required interactions and lower error rates than the traditional system (participant feedback); work featured in NMSU’s Spanish Accessibility Department video of innovative projects

Ciberseguros Lab, New Mexico State University

January 2023 - December 2024

Graduate Research Assistant, Master’s

Las Cruces, New Mexico, USA

- *Collaborators:* Dr. Joshua Reynolds, Manuel Mares Solano, Dr. Bryan White, Dr. Kent Seamons (Brigham Young University), Dr. Tarun Yadav (Brigham Young University)
- Built a first-of-its-kind **deniability** feature in a messaging app and ran a 40+ user study (survey design and pilot interviews) that surfaced key limitations of deniability in secure messaging, achieving a **System Usability Scale** score of 70
- Classified **smart-grid cyberattack** events from news articles using a taxonomy-based framework, analyzing 263 events (124 smart-grid relevant) to characterize threats to critical infrastructure
- Built and maintained an 8-node **Kubernetes** cluster hosting the lab’s models and internal sites, with firewall and privacy controls for shared use

- Advisor: Dr. Smitha Vas
- Built a **Weka**-based **Python** machine-learning app with a team of 4, trained on 1,000+ recruitment questions, that improved mock-interview performance by 20% through actionable feedback
- Achieved 85% accuracy in assessing responses, with iterative refinements improving feedback precision

PUBLICATIONS

Peer-Reviewed

- Leuenberger, J., **Rajendran, A.**, Penzo Jara, A., Hoque, T.U., Darian, S.M. “*Transient Non-Use: How People in Migration Experience Digital Disconnection.*” In Proceedings of CHI 2026.
- **Rajendran, A.**, Yadav, T. K., Al-Jbour, M., Mares Solano, F. M., Seamons, K., & Reynolds, J. (2024, September). “*Deniable Encrypted Messaging: User Understanding After Hands-On Social Experience.*” EuroUSEC Conference 2024, Sweden.
- Aishwarya, P. R., **Rajendran, A.**, Ganga, V. M., Kamarunnisa, K. A. (2019). “*Placement Training using Machine Learning.*” International Journal of Engineering Research & Technology (IJERT), Vol. 8, Issue 6, June 2019.

Manuscripts Under Review

- **Rajendran, A.**, Needle, J., Leuenberger, J., Penzo Jara, A., Hoque, T., Hamilton, W., Darian, S.M. “*Surveillance Perceptions at the US-MX Border.*” Under review at a top-tier ACM venue.
- Needle, J., **Rajendran, A.**, Darian, S.M. “*Privacy Beyond Design: How People on the Move Navigate Visibility and Power.*” Under review at CHI.

Manuscripts in Preparation

- **Rajendran, A.**, Montoya, L., Velarde, R., Cabales, C.G., Hamilton, W., Darian, S.M. “*ClosedCaption: An Offline LLM Transcription/Translation Tool for Qualitative Researchers.*” In preparation.
- Penzo Jara, A., Needle, J., **Rajendran, A.**, Leuenberger, J., Ortiz, G., Darian, S.M. “*Trust, but Verify: Information Seeking and Verification Practices Among People Experiencing Migration.*” In preparation.
- Mares Solano, M., **Rajendran, A.**, Wayllace, C. “*Designing Human-Aware Interfaces: Lessons from LLM Scaffolds in AAC.*” In preparation.

GRANTS & FUNDING

NSF CRII: SaTC (Award No. 2348181), “*Privacy vs. Accountability: Usable Deniability and Non-Repudiation for Encrypted Messaging Systems*” (PI: Dr. William Hamilton). Contributed to the funded proposal; the award supported my M.S. research assistantship and conference travel. Fall 2024

CAHSI-Sloan Grant (Research Capacity): supported research and ethics-in-computing curriculum development, Fall 2025 - Present

PRESENTATIONS

Guest Lectures

- “Ethics in Generative AI”, CSCI 5440: Generative Artificial Intelligence (Dr. Son Tran), Spring 2026
- “Ethics in Deep Learning”, CSCI 4425: Introduction to Deep Learning (Dr. Huping Cao), Fall 2025

Conference & Research Talks

- “Enhancing AAC Communication with Language Models”, CS AI Monthly Meetups (Dr. Christabel Wayllace), Spring 2025

- “Deniable Encrypted Messaging: User Understanding After Hands-On Social Experience”, EuroUSEC 2024, Sweden

Posters

- “Deniable Encrypted Messaging”, NMSU Graduate Research Week, Fall 2024

TEACHING EXPERIENCE

Ethics in Computing Curriculum Developer, CAHSI

New Mexico State University

2025 - Present

Las Cruces, New Mexico, USA

- *Collaborators:* Dr. Shiva Darian, Dr. Bill Hamilton, Dr. Enrico Pontelli (Dean of Arts and Sciences), Raena Cota (CAHSI Connector, Southwest Region), Dr. Timothy Cleveland (Department Head, Philosophy), Zimzi Lovett (Department of Philosophy)
- Designing an embedded ethics-in-computing curriculum (modeled on Harvard’s Embedded EthiCS program) that integrates ethics modules into individual CS courses
- Successfully implemented it in CSCI 3710: Software Development (Dr. Bill Hamilton), delivering three tiered assignments (a Moral Machine exercise, a coding assignment, and a role-playing game) to 40 to 50 students

Teaching Assistant, Department of Computer Science

New Mexico State University

Spring 2024

Las Cruces, New Mexico, USA

- Teaching Assistant for CSCI 4245: Computer Networks I and CSCI 4250: Human-Centered Computing (Dr. Joshua Reynolds), each enrolling 40 to 50 graduate and undergraduate students
- Supported students through assignments, lab sessions, and grading; received departmental recognition for teaching quality

MENTORING

Application & Fellowship Mentor, CAHSI

New Mexico State University

Summers 2025 & 2026; Fall 2025 & Fall 2026

Las Cruces, New Mexico, USA

- Mentored undergraduate and graduate students on graduate school and fellowship applications (e.g., NSF GRFP), giving feedback on resumes, personal statements, and application materials

Research & Software Development Mentor

New Mexico State University

2025

Las Cruces, New Mexico, USA

- Mentored Bryan White, a Psychology PhD student (now an Assistant Professor in NMSU’s Psychology Department), building the software for his dissertation study: a JATOS-hosted jsPsych online experiment (“SpamSim”) in HTML, CSS, and JavaScript (Fall 2025)
- Mentored Lucas Montoya, Rayannon Velarde, and Christian Gabriel Cabales on their senior capstone project, guiding them to build the graphical interface for my WhisperX-based translation tool (ClosedCaption)
- Mentored undergraduate researcher Abdulhakim Abdulsalam on analyzing and enhancing the AAC word-prediction component as part of a summer undergraduate research program

LEADERSHIP EXPERIENCE

Co-Founder & Vice President, NMSU International Collaborative Coding Club August 2023 - May 2025

New Mexico State University

Las Cruces, New Mexico, USA

- Co-founded the student organization and served as Vice President, managing the team, meetings, and project workflows
- Led development of the club’s text-to-speech communication software (see Tools Built), integrating the WASI Lab AAC word-prediction tool into a user-facing GUI for broader testing

Treasurer, Computer Science Graduate Student Organization (CSGSO) August 2023 - January 2025

New Mexico State University

Las Cruces, New Mexico, USA

- Managed organization finances and budget allocation for graduate student activities
- Coordinated travel funding and supported graduate students navigating funding requests and graduate-level processes

WORK EXPERIENCE

Infosys Limited

October 2019 - September 2022

Full Stack Developer

Trivandrum, Kerala, India

- Led a team of 4 UI developers to build a responsive Angular web application for FedEx's logistics service, increasing customer satisfaction and engagement by roughly 30%
- Implemented Spring Boot back-end services that cut average page-load times by 5 seconds and improved overall site speed by 20%
- Automated testing (Jasmine) and CI/CD (Jenkins) to maintain a 95% bug-free codebase and cut time-to-market by 30%, delivering via agile methodologies

TOOLS BUILT

ClosedCaption: Local Transcription & Diarization Tool

May 2025 - Present

DAR Lab, NMSU

- On-device transcription and speaker-diarization pipeline (WhisperX) built for confidential audio; in use on 100+ Spanish-language interviews across three research projects

LLM-Assisted Single-Pointing AAC Keyboard

January 2025 - June 2025

WASI Lab, NMSU

- On-screen single-pointing keyboard that uses an LLM to predict words and sentences for non-verbal users, reducing required interactions and error rates versus a traditional joystick-based AAC system

ALoud: Text-to-Speech Communication App

May 2023 - May 2025

NMSU International Collaborative Coding Club

- Full text-to-speech app (ReactJS, Chakra UI, TypeScript) with word prediction, an acronym dictionary, adjustable speech speed, and multi-language support
- Added expressive audio: preset sounds for common phrases and emotive cues (sigh, disgust, happiness, questioning, laughter)

SKILLS AND INTERESTS

Research Interests	Usable Security and Privacy, Human-Computer Interaction, Augmentative and Alternative Communication (AAC), Qualitative Research, Smart Grid Security, Ethics in Computing
Methods & Tools	Qualitative & Quantitative Analysis, NVivo, Mural, WhisperX, Ollama, Python, LaTeX
Programming	Python, Java, TypeScript, HTML, CSS, SQL, NoSQL (MongoDB)
Tools & Frameworks	Angular, React, ChakraUI, Bootstrap, Wireshark, AWS EC2, AWS S3, Hadoop, Kubernetes

ACHIEVEMENTS

Finalist, Lucy Parsons Labs Summer Fellowship, 2026 (among 200+ applicants worldwide)

Selected as **Student Volunteer** for CHI 2026, Spain

Presented at **EuroUSEC 2024** (Sweden) as the only Master's student among PhD-level presenters

Best TA Award, Department of Computer Science, NMSU, Fall 2023

Second Prize, 3-Minute Lightning Talk among 20 participants, Research Creativity Week, NMSU

Second Prize, Oral Presentation among 35 participants, Research Creativity Week, NMSU

Infosys INSTA Award of Excellence for rapid delivery and automation of projects for two consecutive years

FedEx-Infosys Certificate of Excellence for contributions to the FedEx ROCS application as a developer

Infosys Bronze Medal for high code quality, 90% coverage, and automated deployments in an engineering project

Ranked first in the Computer Science department for three consecutive years during B.Tech; awarded an INR 10,000 merit scholarship

CERTIFICATIONS

CITI Program: Social & Behavioral Research (Refresher) and Responsible Conduct of Research (Refresher), Apr 2026 (valid through Apr 2029)

AWS Certified Cloud Practitioner, Amazon Web Services (Mar 2023)

Infosys Certified Agile Developer, Infosys Ltd. (Sep 2020)

Infosys Certified Front End Developer, Infosys Ltd. (Aug 2020)

Machine Learning, Harvard University (Aug 2020)